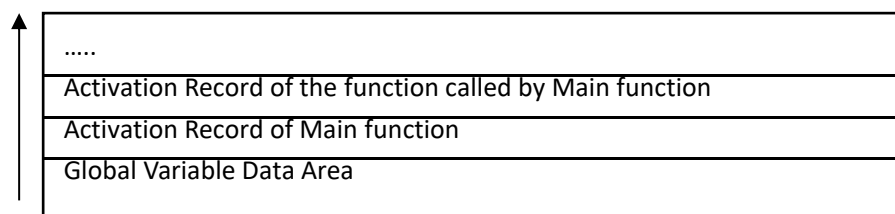


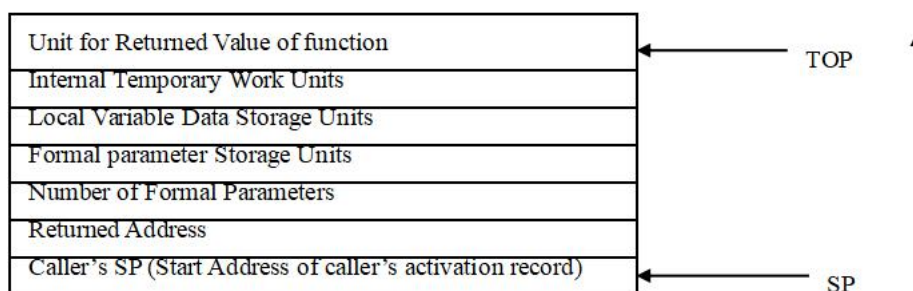
Written Assignment 6- run-time environment management

1. Please explain the following terms in your own language
 - (1) Program and Process
 - (2) Activation tree and Control stack
 - (3) Environment and State
2. Please give the general map of the data area of a process, the general map of a function's record. A detailed introduction of the maps is needed.
3. The storage organization and the form of activation record used in C language are as following. Please construct the run-time stack map when it gets the maximum size for the first time for the following C program.

Storage Organization of C Language



The Form of Activation Record of any a function in C



The C program is as the following:

```
#include <stdio.h>
```

```

int x,y;
int main()
{ x=5; y=f(x);}
int f(int n)
{ if (n<=1)
    return 1;
  else
    if ( n==2) return 2;
  else
    { int t1,t2,t3,t4,t;
      t1=f(n-1); t2=f(n-2); t3=f(n-3); t4=t1+t2;t=t3+t4; return t
    }
}

```

Notes: 1) Here we assume that the caller's sp of Main function is the start address of global variable data area, and the returned address in the activation record of a function (including Main function) is filled by the operating system automatically, you might not care it.

2) The initial value of variable X is 5, the start address of stack used in the program is K.

3) The stack map may get its maximum size for several times, here we ask you draw the stack map at maximum size for the first time.

4. Please construct the run-time stack map (detailed map) when it gets

the maximum size for the second time for the following C program

```

#include <stdio.h>

int x,y,z;

int main()
{
    x=5;
    y=8;
    z=f(x,y);
}

int f(int m, int n)
{
    if (m>=0 && n>=0)
    {
        if (m==0) return n;
        else if (n==0) return m;
    }
}

```

```

else
{
    int t1,t2,t;
    if (m>=n) {
        t1=m-n;
        t2=n
    }
    else {
        t1=n-m;
        t2=m
    }
    t=f(t1,t2);
    return t
}
}

```

Notes: 1) Here we assume that the caller's sp of Main function is the start address of global variable data area, and the returned address in the activation record of a function (including Main function) is filled by the operating system automatically, you might not care it.

2) The start address of stack used in the program is K.

Complete the homework and submit it before the next next class to the teaching assistant (Room 156, Building of Sch. Computer Sci&Eng).

1. Everyone should give their completed workbook to the teaching assistant before the designated time. There is a box at the door of the room specifically for collecting homework.
2. After the first submission of each assignment, it will be graded based on the quality of completion. Allow the next submission to modify errors from the previous assignment. The final score for each assignment will be given based on the final quality and number of revisions made.